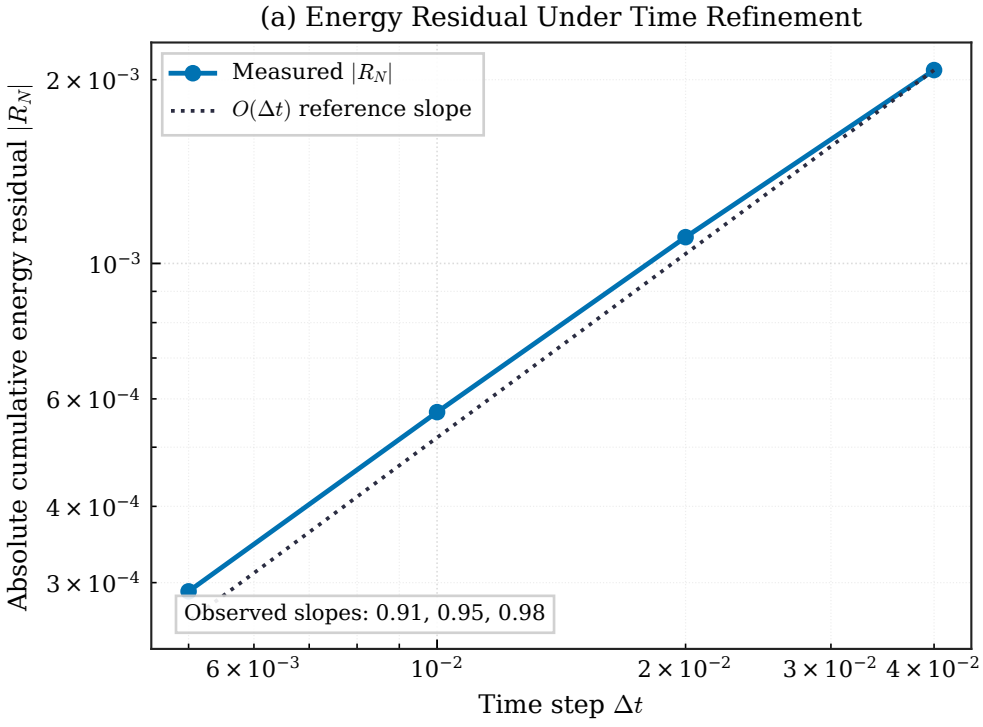
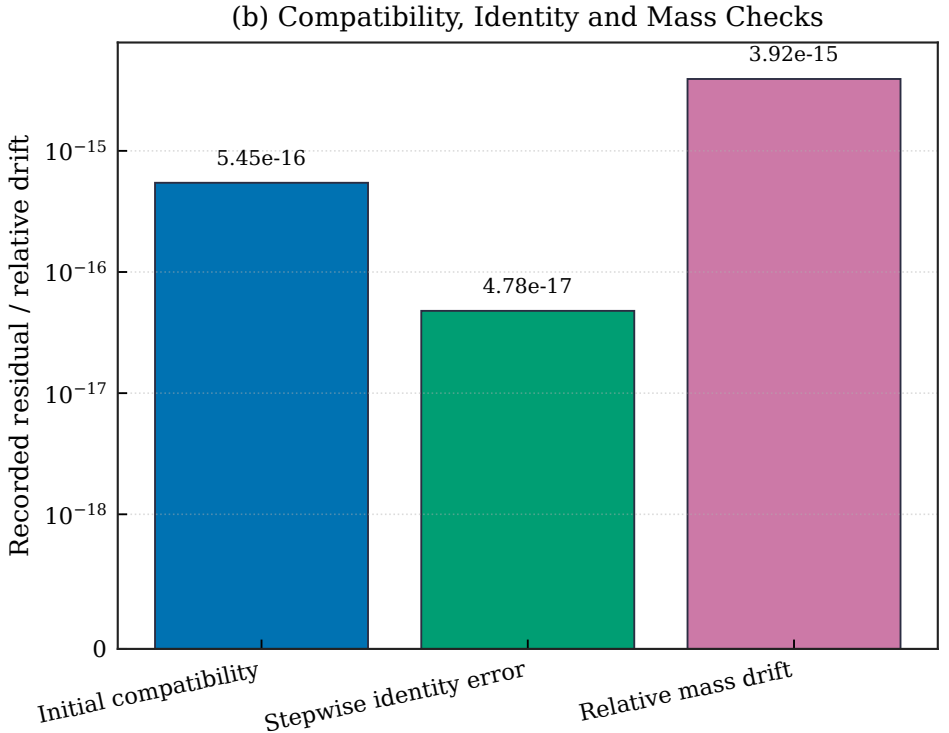

(a) Energy Residual Under Time Refinement
Measured $|R_N|$
$O(\Delta t)$ reference slope
Observed slopes: 0.91, 0.95, 0.98
Absolute cumulative energy residual $|R_N|$
Time step Δt
2×10^{-3}
10^{-3}
6×10^{-4}
4×10^{-4}
3×10^{-4}
6×10^{-3}
10^{-2}
2×10^{-2}
3×10^{-2}
4×10^{-2}


(b) Compatibility, Identity and Mass Checks
Recorded residual / relative drift
5.45e-16
4.78e-17
3.92e-15
10^{-15}
10^{-16}
10^{-17}
10^{-18}
0
Initial compatibility
Stepwise identity error
Relative mass drift